

Lecture 21: Curl and Divergence

Background & Motivation

Green's Theorem involved the quantity $Q_x - P_y$. Today we give it a name: the **curl**. We also introduce the **divergence**, which measures expansion or compression. These two differential operators are the vocabulary of the final theorems of the course (Stokes' and the Divergence Theorem).

1. Curl: Measuring Rotation

Definition 1: Curl of a Vector Field

For $\mathbf{F} = \langle P, Q, R \rangle$, the **curl** is (all the same thing, different notation):

$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F} = \nabla \times \mathbf{F} = \langle R_y - Q_z, P_z - R_x, Q_x - P_y \rangle$$

How to compute (component by component):

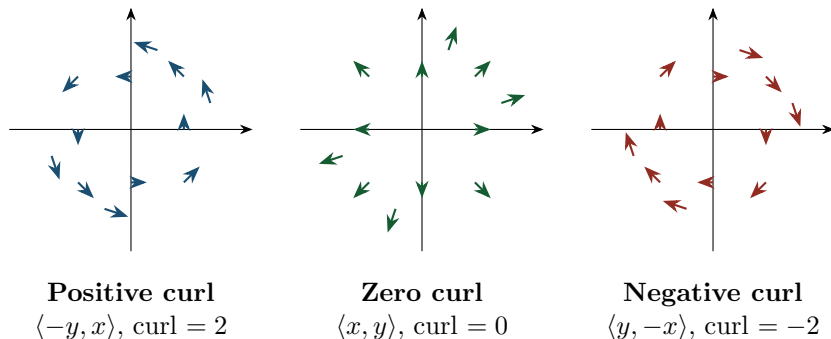
$$\bullet \hat{\mathbf{i}}: \frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \quad \hat{\mathbf{j}}: \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \quad \hat{\mathbf{k}}: \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$$

Memory trick: each component uses the two variables it doesn't point along, in cyclic order ($x \rightarrow y \rightarrow z \rightarrow x \rightarrow \dots$). Always subtract in "reverse" order. If you know determinants, this equals

$$\begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}, \text{ but that is not required.}$$

- **Interpretation:** measures the **local rotation** ("swirling") of the field at a point
- $\nabla \times \mathbf{F} = \mathbf{0} \implies$ **irrotational** (no spinning)
- Direction of $\nabla \times \mathbf{F}$ = axis of rotation (right-hand rule), magnitude = rate of rotation

Visualizing Curl



The Paddlewheel Test

Think of placing a tiny paddwheel at a point in the flow. The fluid pushes on each blade. If the flow is faster or differently directed on one side than the other, the imbalance creates a torque and the wheel spins.

- **Positive curl** ($\nabla \times \mathbf{F} > 0$ in 2D): paddwheel spins **CCW**. Right-hand rule: thumb out of page ($+\hat{\mathbf{k}}$).
- **Negative curl** ($\nabla \times \mathbf{F} < 0$ in 2D): paddwheel spins **CW**. Right-hand rule: thumb into page ($-\hat{\mathbf{k}}$).
- **Zero curl**: paddwheel doesn't spin, even if fluid is moving fast.

Key insight: curl measures *local spin*, not global shape. A field can have curved streamlines but zero curl, and *straight* streamlines but nonzero curl!

Example — shear flow: In $\mathbf{F} = \langle y, 0 \rangle$ every arrow is horizontal, yet the paddwheel spins. The top blade ($y > 0$) is pushed right; the bottom blade ($y < 0$) is pushed left. This imbalance $\implies$ **CW** rotation, and indeed $Q_x - P_y = 0 - 1 = -1 < 0$. Negative curl despite no circular streamlines.

In 3D, $\nabla \times \mathbf{F}$ is a vector: its **direction** is the axis of fastest rotation (right-hand rule) and its **magnitude** is the rate of rotation.

Example 1: Determining Curl Sign from the Paddwheel

For each field, predict the paddwheel behavior at the origin, then verify by computing the 2D curl.

- (a) $\mathbf{F} = \langle 0, x \rangle$
- (b) $\mathbf{F} = \langle -y, 0 \rangle$
- (c) $\mathbf{F} = \langle y, x \rangle$

(a) Right of origin ($x > 0$): upward push. Left ($x < 0$): downward push. $\implies$ **CCW**.
 $Q_x - P_y = 1 - 0 = 1 > 0$ ✓ Positive curl (CCW).

(b) Above origin ($y > 0$): leftward push. Below ($y < 0$): rightward push. $\implies$ **CCW**.
 $Q_x - P_y = 0 - (-1) = 1 > 0$ ✓ Positive curl (CCW). Different-looking field, same curl as (a)!

(c) From $P = y$: top-right, bottom-left (CW tendency). From $Q = x$: right-up, left-down (CCW tendency). They **cancel**.
 $Q_x - P_y = 1 - 1 = 0$ ✓ Zero curl. In fact $\mathbf{F} = \nabla(xy)$ — conservative.

2D Reduction

For $\mathbf{F} = \langle P, Q \rangle$ (2D, so $R = 0$), only the $\hat{\mathbf{k}}$ component survives: $\nabla \times \mathbf{F} = (Q_x - P_y)\hat{\mathbf{k}}$. We call the scalar $Q_x - P_y$ the **2D curl** — exactly the integrand in Green's Theorem!

Example 2: 3D Curl Computation

Find $\nabla \times \mathbf{F}$ for $\mathbf{F} = \langle xz, xyz, -y^2 \rangle$.

$$P = xz, \quad Q = xyz, \quad R = -y^2.$$

$$\hat{\mathbf{i}}: R_y - Q_z = (-y^2)_y - (xyz)_z = -2y - xy$$

$$\hat{\mathbf{j}}: P_z - R_x = (xz)_z - (-y^2)_x = x - 0 = x$$

$$\hat{\mathbf{k}}: Q_x - P_y = (xyz)_x - (xz)_y = yz - 0 = yz$$

$$\nabla \times \mathbf{F} = \langle -2y - xy, \quad x, \quad yz \rangle$$

Example 3: 2D Curl: Rotation vs. Source

Compute the 2D curl of (a) $\mathbf{F} = \langle -y, x \rangle$ and (b) $\mathbf{F} = \langle x, y \rangle$.

(a) $Q_x - P_y = (x)_x - (-y)_y = 1 - (-1) = 2$. Positive $\implies$ CCW rotation everywhere.

(b) $Q_x - P_y = (y)_x - (x)_y = 0 - 0 = 0$. Irrotational! Despite pointing outward everywhere, there is no local spinning. (This field is conservative: $\mathbf{F} = \nabla\left(\frac{x^2+y^2}{2}\right)$.)

2. Divergence: Measuring Expansion

Definition 2: Divergence of a Vector Field

For $\mathbf{F} = \langle P, Q, R \rangle$, the **divergence** is (same thing, different notation):

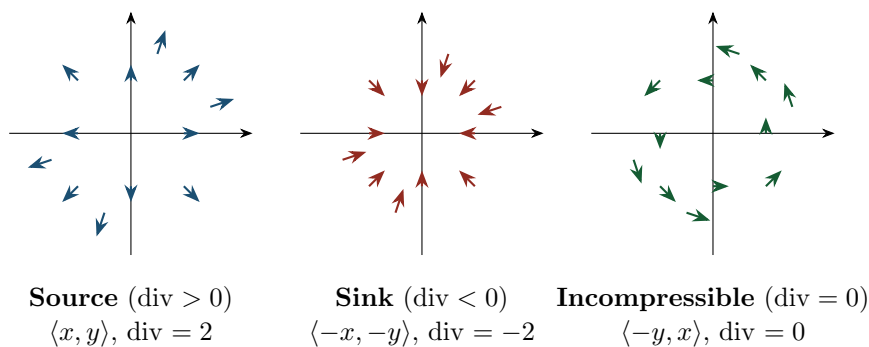
$$\operatorname{div} \mathbf{F} = \operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

Just add up how much each component changes along its own direction.

– **Interpretation:** measures the net “**outflow**” or **expansion at a point**

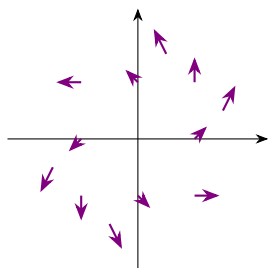
– $\operatorname{div} \mathbf{F} > 0$: **source** (fluid spreading out) $\operatorname{div} \mathbf{F} < 0$: **sink** (fluid compressing) $\operatorname{div} \mathbf{F} = 0$: **incompressible**

Visualizing Divergence

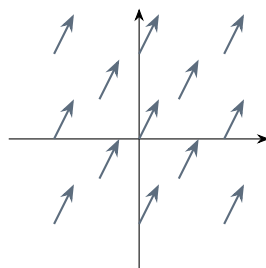


Both Properties / Neither

Curl and divergence are **independent**. A field can have any combination:



Both ($\text{curl} \neq 0, \text{div} \neq 0$)
 $\langle x - y, x + y \rangle$
 $\text{curl} = 2, \text{div} = 2$



Neither ($\text{curl} = 0, \text{div} = 0$)
 $\langle 1, 2 \rangle$
 Uniform flow

Example 4: Incompressible Pipe Flow

A velocity field in a pipe is $\mathbf{F} = \langle 2x, -y, -z \rangle$. Is the flow incompressible?

$$\text{div } \mathbf{F} = (2x)_x + (-y)_y + (-z)_z = 2 + (-1) + (-1) = 0$$

Yes — $\text{div } \mathbf{F} = 0$, so the flow is **incompressible**. The fluid speeds up along x (pipe narrows) but compensates by slowing in y and z . No fluid is created or destroyed.

Example 5: Classifying a Field

For $\mathbf{F} = \langle x, y, z \rangle$, find $\text{div } \mathbf{F}$ and $\nabla \times \mathbf{F}$. Interpret.

$$\text{div } \mathbf{F} = 1 + 1 + 1 = 3 \quad (\text{source everywhere — uniform expansion})$$

$$\nabla \times \mathbf{F}: \quad \hat{\mathbf{i}}: 0 - 0 = 0, \quad \hat{\mathbf{j}}: 0 - 0 = 0, \quad \hat{\mathbf{k}}: 0 - 0 = 0. \quad \text{So } \nabla \times \mathbf{F} = \mathbf{0} \text{ (irrotational).}$$

Like an explosion: everything moves outward, nothing rotates.

Example 6: Another Classification

For $\mathbf{F} = \langle -y, x, 0 \rangle$, find $\text{div } \mathbf{F}$ and $\nabla \times \mathbf{F}$. Interpret.

$$\text{div } \mathbf{F} = (-y)_x + (x)_y + (0)_z = 0 \quad (\text{incompressible})$$

$$\nabla \times \mathbf{F}: \quad \hat{\mathbf{i}}: 0 - 0 = 0, \quad \hat{\mathbf{j}}: 0 - 0 = 0, \quad \hat{\mathbf{k}}: 1 - (-1) = 2. \quad \text{So } \nabla \times \mathbf{F} = 2\hat{\mathbf{k}} \text{ (CCW about } z\text{).}$$

Like a merry-go-round: everything spins, nothing expands or compresses.

3. Key Identities

Vector Calculus Identities

1. $\nabla \times (\nabla f) = \mathbf{0}$ — conservative fields are always irrotational (by Clairaut's: $f_{xy} = f_{yx}$)

Contrapositive: if $\nabla \times \mathbf{F} \neq \mathbf{0}$, then $\mathbf{F}$ is **not** conservative.

2. $\text{div}(\nabla \times \mathbf{F}) = 0$ — curl fields are always divergence-free (Clairaut's again)

3. **Laplacian:** $\nabla^2 f = \text{div}(\nabla f) = f_{xx} + f_{yy} + f_{zz}$ — divergence of the gradient

4. **Product rules:**

$$\nabla(fg) = f\nabla g + g\nabla f \quad \text{div}(f\mathbf{F}) = f \text{div } \mathbf{F} + \mathbf{F} \cdot \nabla f \quad \nabla \times (f\mathbf{F}) = f \nabla \times \mathbf{F} + \nabla f \times \mathbf{F}$$

Operator Relationships

$$\text{Scalar field } f \xrightarrow{\text{grad}} \text{Vector field } \mathbf{F} \xrightarrow{\text{curl}} \text{Vector field } \mathbf{G} \xrightarrow{\text{div}} \text{Scalar field } h$$

The two zero identities say: $\text{grad} \rightarrow \text{curl} = \mathbf{0}$, and $\text{curl} \rightarrow \text{div} = 0$. In other words, the “derivative of a derivative” is zero in this chain.

Example 7: Verifying $\nabla \times (\nabla f) = \mathbf{0}$

Let $f(x, y, z) = x^2yz + \sin(yz)$. Verify $\nabla \times (\nabla f) = \mathbf{0}$.

Gradient: $\nabla f = \langle 2xyz, x^2z + z \cos(yz), x^2y + y \cos(yz) \rangle$

Curl:

$\hat{\mathbf{i}}$: $(x^2y + y \cos(yz))_y - (x^2z + z \cos(yz))_z = [x^2 + \cos(yz) - yz \sin(yz)] - [x^2 + \cos(yz) - yz \sin(yz)] = 0$

✓

$\hat{\mathbf{j}}$: $(2xyz)_z - (x^2y + y \cos(yz))_x = 2xy - 2xy = 0$ ✓

$\hat{\mathbf{k}}$: $(x^2z + z \cos(yz))_x - (2xyz)_y = 2xz - 2xz = 0$ ✓

This always works by Clairaut’s theorem ($f_{xy} = f_{yx}$, etc.).

Example 8: Proving $\text{div}(\nabla \times \mathbf{F}) = 0$

Show that $\text{div}(\nabla \times \mathbf{F}) = 0$ for any smooth $\mathbf{F} = \langle P, Q, R \rangle$.

$\nabla \times \mathbf{F} = \langle R_y - Q_z, P_z - R_x, Q_x - P_y \rangle$

$\text{div}(\nabla \times \mathbf{F}) = (R_y - Q_z)_x + (P_z - R_x)_y + (Q_x - P_y)_z$

$= R_{yx} - Q_{zx} + P_{zy} - R_{xy} + Q_{xz} - P_{yz}$

By Clairaut’s: $R_{yx} = R_{xy}$, $Q_{zx} = Q_{xz}$, $P_{zy} = P_{yz}$. Every term cancels. $\square$

4. Green’s Theorem Reformulated

Green’s Theorem via Curl

$$\int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \iint_D (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{k}} dA$$

The circulation of $\mathbf{F}$ around C equals the total curl over D . This is the 2D special case of **Stokes’ Theorem** (coming soon!).

Practice Problems

Problem 1. Compute $\text{div } \mathbf{F}$ and $\nabla \times \mathbf{F}$ for $\mathbf{F} = \langle x^2, y^2, z^2 \rangle$.

Answer: $\text{div } \mathbf{F} = 2x + 2y + 2z$, $\nabla \times \mathbf{F} = \mathbf{0}$

Problem 2. Show that $\mathbf{F} = \langle yz, xz, xy \rangle$ is both irrotational and incompressible.

Answer: $\nabla \times \mathbf{F} = \mathbf{0}$, $\text{div } \mathbf{F} = 0$

Problem 3. Find $\nabla \times \mathbf{F}$ and $\operatorname{div} \mathbf{F}$ for $\mathbf{F} = \langle y^2z, x^2z, x^2y \rangle$.

Answer: $\nabla \times \mathbf{F} = \langle 0, y^2 - 2xy, 2z(x - y) \rangle$, $\operatorname{div} \mathbf{F} = 0$

Problem 4. Compute $\nabla \times \mathbf{F}$ for $\mathbf{F} = \langle e^x \cos y, -e^x \sin y, 0 \rangle$. What does the result tell you?

Answer: $\nabla \times \mathbf{F} = \mathbf{0} \implies$ conservative, potential $f = e^x \cos y$

Problem 5. Verify $\operatorname{div}(f\mathbf{F}) = f \operatorname{div} \mathbf{F} + \mathbf{F} \cdot \nabla f$ for $f = xy$ and $\mathbf{F} = \langle 1, 1, 0 \rangle$.

Answer: both sides $= x + y$ ✓

Problem 6. Prove: if $\mathbf{F} = \nabla f$ and f is harmonic ($\nabla^2 f = 0$), then $\mathbf{F}$ is both irrotational and incompressible.

Irrotational: $\nabla \times \mathbf{F} = \nabla \times (\nabla f) = \mathbf{0}$ (identity 1).

Incompressible: $\operatorname{div} \mathbf{F} = \operatorname{div}(\nabla f) = \nabla^2 f = 0$ (since f is harmonic). $\square$

Problem 7. Without computing, predict whether the 2D curl is positive, negative, or zero at the origin. Then verify.

(a) $\mathbf{F} = \langle -2y, 3x \rangle$ (b) $\mathbf{F} = \langle y, y \rangle$ (c) $\mathbf{F} = \langle x^2, -y^2 \rangle$

Answer: (a) $Q_x - P_y = 3 - (-2) = 5$ (positive, CCW). (b) $Q_x - P_y = 0 - 1 = -1$ (negative, CW).

(c) $Q_x - P_y = 0$ (irrotational).

Problem 8. The velocity field of a whirlpool is modeled by $\mathbf{F} = \langle -y, x, 2z \rangle$. Find $\nabla \times \mathbf{F}$ and $\operatorname{div} \mathbf{F}$. Does this flow rotate, expand, or both?

$\nabla \times \mathbf{F} = \langle 0 - 0, 0 - 0, 1 - (-1) \rangle = 2\hat{\mathbf{k}}$ (CCW rotation about z -axis)

$\operatorname{div} \mathbf{F} = 0 + 0 + 2 = 2$ (expanding vertically)

Both! Like a tornado: spinning while air is pulled upward.